

quickly the core technology can be scaled in terms of the number of beams and beam current.

Meanwhile, metrology, the science of measuring structures and films, is another concern. Today's metrology tools are capable of measuring structures in two dimensions, and in three dimensions to some degree. But that is not nearly enough for the complexity of current and future devices. In fact, there is no single metrology system that can measure everything. So chipmakers must use several different metrology tools in the flow.

As the industry moves to 7nm and 5nm, there will be a drastic decline in signal-to-noise ratio of metrology tools. To compensate this uncertainty of measurements from each tool, there would be an even greater need for a multiple metrology approach. This means hybrid metrology is inevitable.

In hybrid metrology, chipmakers use a mix and match of several different tool technologies and then combine data from each. Hybrid metrology, however, is still in developing stage. Fortunately, this silicon alternative does not have any major physical flaw like most other materials.

Silicene and germanene, for example, degrade within a few minutes or hours, while graphene is not suitable for all transistor types and is still extremely challenging to use in the production of integrated circuits due to fragility of its structure. InGaAs and InP based devices can be manufactured using existing techniques in combination with silicon.

The main issue remains to be yield and cost. Beyond 14nm, as we move to 10nm and 7nm, a new fin material will be required—probably silicon-germanium (SiGe), or perhaps just pure germanium. SiGe and Ge have higher electron mobility than Si, allowing

for lower voltages and, thus, reducing power consumption, tunneling and leakage.

SiGe has been used in commercial CMOS fabrication since the late 1980s, too, so switching from silicon would not be too painful. The primary reason that we have been using silicon for so long is that the entire industry is based on silicon. The amount of time, money and R&D that would be required to deploy new machines for handling new materials that we know relatively little about would be astronomical.

SiGe would take us to 7nm, but after that we are probably looking at a new transistor structure. Just as FinFET created a larger surface area, mitigating the effects of quantum tunneling, both gate-all-around FETs and vertical tunneling FETs would again allow for shorter gates and lower voltages.

Gate-all-around FET essentially consists of nanowire source and drains, surrounded by a gate. A vertical tunneling FET is similar in a way that it uses nanowires, but the actual method of operation is very different from conventional FETs. TFETs allow for lower operating voltage.

Another option is a somewhat conventional FinFET, but with fin constructed out of III-V semiconductors such as gallium-arsenide (GaAs), which again have higher electron mobility than silicon. There may be better gains to be had from moving sideways, to materials and architectures that can operate at faster frequencies and with more parallelism, rather than brute-forcing the continuation of Moore's law.

Meaning, the pitch or distance between major features on the chip are still scaling at an expected rate. Reduction in distance between fins, the raised channels behind the name FinFETs, is right on target (0.70x), the distance between gates

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